

# ADITYA LAWANDE

Full-Stack Software Engineer | Node.js, Next.js, PostgreSQL, LLM Integration  
+91 8605485019 | adityalawande1771@gmail.com | LinkedIn | GitHub | Portfolio

## EXPERIENCE

### Software Engineer Intern

June 2026 – August 2026

*Spherical Insights LLP, Pune*

*On-site*

- Engineered a full-stack content publishing CMS solo using Java Spring Boot, Thymeleaf and MySQL, unifying **7 content types** (blogs, news, interviews, case studies and more) under one content-type-driven data model.
- Developed frontend pages and backend workflows for market research portals, connecting UI components to server-side logic and relational queries.
- Co-built a multilingual report filtering feature supporting database-driven content discovery across **14 languages**.

## PROJECTS

**SPPU Study Hub** | sppustudyhub.in | GitHub | *Node.js, PostgreSQL, pgvector, Claude API, OpenAI Embeddings, Cloudflare R2*

- Designed two independent auth systems (JWKS/ES256 OAuth for users, a separate JWT scheme for admins) with all authorization enforced server-side from verified token identity; platform serves **6,000+ monthly visitors, 36,000+ monthly page views** and **750+ registered users**.
- Traced a production login failure to a JWKS/shared-secret mismatch that rejected valid sessions despite a fully passing test suite; pinpointed a test-coverage gap rather than app logic and corrected the token verification flow.
- Cut dashboard requests per page load from **~20 to 5** by replacing per-subject API fan-out (N+1) with one batched SQL endpoint, reducing concurrent DB queries **fivefold**.
- Integrated Claude API for AI exam explanations with semantic embedding-based response caching (OpenAI Embeddings, pgvector), reducing repeated LLM calls by **65–70%** under real usage limits.

**Studentkart** | Live | GitHub | *Next.js, PostgreSQL/PostGIS, Prisma, Supabase, Socket.IO*

- Implemented geospatial listing search on PostgreSQL/PostGIS with a GIST-indexed geography column and raw SQL, plus a live /benchmark page comparing it against a naive in-app filter across **5,000 seeded listings**.
- Deployed a standalone Node.js/Socket.IO chat service on Render, separate from the Next.js app on Vercel, authenticating socket handshakes with short-lived JWTs minted by the main app.
- Diagnosed a production Prisma P1001 failure across Vercel, Render and Supabase as IPv6-only reachability from Vercel's serverless runtime; switched to a pooled connection endpoint and confirmed the fix on live traffic.

**Farmsense AI** | Live | GitHub | *React.js, Node.js, Express.js, MongoDB, Claude API, RAG*

- Architected an agentic advisory system on Claude's native tool-use API, combining live weather, mandi price trends, farm records and soil requirements into one contextual recommendation.
- Grounded advice in retrieved ICAR/KVK documents with citations through a RAG pipeline, evaluated with **~20 deterministic test cases** plus LLM-as-judge scoring to curb hallucinated regional advice.
- Lowered LLM spend by enabling Anthropic prompt caching (validated via `cache_read/cache_creation` metrics after diagnosing model-specific eligibility) and fixing a React effect-dependency bug that fired duplicate API calls.

## TECHNICAL SKILLS

**Languages:** JavaScript, SQL, Java, Python, C++

**Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS

**Backend:** Node.js, Express.js, Spring Boot, REST APIs, Socket.IO, Server-Sent Events

**Databases:** PostgreSQL, PostGIS, MySQL, MongoDB, Prisma, Supabase, pgvector

**Auth & Cloud:** JWT, OAuth 2.0 / JWKS, Supabase Auth, Vercel, Render, Cloudflare R2, Git, Postman

**AI:** Claude API (tool use, prompt caching), OpenAI Embeddings, RAG

**Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## EDUCATION

**JSPM's Jayawantrao Sawant College Of Engineering, Pune**

Aug 2023 – June 2027

*Bachelor of Engineering in Information Technology | CGPA: 8.28*

## ACHIEVEMENTS

**LeetCode:** Solved **250+** problems across arrays, linked lists, trees, graphs and dynamic programming.